PERSONAL LEARNING JOURNEY

IT3091 - Machine Learning | Bank Marketing Project

Student ID: IT24101461 **Role:** Member 2 - Feature Engineering and Preprocessing

1. My contribution

- I prepared features for predicting term-deposit subscriptions and prioritising marketing contacts. I copied Member 1's training/test splits to preserve the original data.
- I created eight features, assembled preprocessing, selected 30 features and saved the outputs for the next modelling stage.

2. Feature decisions and reasons

Features created	Decision and purpose
previously_contacted; pdays_clean	Replace pdays = -1 with 0 in the cleaned feature; retain a contact indicator so the original meaning is preserved.
loan_count; has_credit_default	Count housing/personal loan indicators; represent credit default separately because repayment status is different.
previous_campaign_success	Represent a successful previous campaign outcome as a binary indicator.
age_group; balance_group; campaign_contact_group	Group age, balance and contact effort to represent possible response patterns; retain the original variables.

3. Preprocessing and feature selection

- **Numerical:** median imputation and StandardScaler. **Categorical:** most-frequent imputation and OneHotEncoder with unseen-category handling, combined in a ColumnTransformer.
- I fitted preprocessing and selection on training data and only transformed test data. SelectKBest with mutual_info_classif retained 30 features, scoring dependence that may be non-linear.

4. Verified outputs and handover

Stage / check	Training set	Test set
Original inputs	36,168 rows x 15 columns	9,043 rows x 15 columns
After feature creation	23 columns	23 columns
After preprocessing / selection	68 features / 30 selected	68 features / 30 selected
Validation	No missing values; columns match	No missing values; columns match

I saved three files with the **IT24101461_** prefix: **engineered_data.joblib** (engineered splits), **selected_data.joblib** (selected arrays, targets and feature names), and **feature_pipeline.joblib** (fitted preprocessing and selector).

5. Challenges, learning and future improvements

- **Meaning and consistency:** handling pdays taught me why a special value needs interpretation. Checking shapes, missing values and column order helped verify a consistent team handover.
- **Evidence-based decisions:** fewer features do not prove better prediction. I used mutual information instead of the proposed ANOVA method, but did not establish that it was better or that 30 features was optimal.
- **Next improvements:** fit preprocessing and selection within each cross-validation fold; add feature creation to the saved pipeline, which currently expects engineered columns; compare feature counts, set a mutual-information random seed and test the full raw-input handover.

Evidence: IT24101461_FeatureEngineering.ipynb and its three saved joblib outputs.
AI-use transparency: ChatGPT assisted with drafting and formatting from my supplied notebook and project documents.